

Exploring Customer Behavior

A Comprehensive SQL Case Study on Database Analysis &
Strategic Insight for Online Food Retailers

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Unlocking The Narrative of Data

As a Data Analyst, the most compelling task is transforming silent rows of data into a narrative that explains **human behavior**. This project decodes visiting patterns and spending habits to build a deeper connection between brands and customers.



Engagement Pattern Analysis



Menu Preference Profiling



Loyalty Impact Evaluation

Data Analytics Dashboard

Defining Analytical Objectives

Behavioral Analysis

Track visit frequencies, identify customer entry points through "first orders," and determine menu popularity between standard and gold members.

Financial Optimization

Calculate revenue contribution per product, determine Customer Lifetime Value (CLV), and analyze annual growth trends to support operational efficiency.

Database Schema

The system is built on 5 core integrated tables containing 62 transactional rows and dimensional attributes.

Fact Tables:

 Sales: Transaction records

Dimension Tables:

 Product: Catalog & Pricing

 Users: Registration master

 Golduser_Signup: Loyalty tracking

| The SQL Methodology



Data Aggregation: Leveraging COUNT(), SUM(), and AVG() to summarize workforce and transaction metrics.



Advanced Querying: Utilizing Common Table Expressions (CTEs) and Window Functions like RANK() to identify top-performing segments.



Data Transformation: Implementing CASE statements and complex JOIN logic to simulate behavior shifts and membership perks.

Product Portfolio Revenue



Butter Chicken dominates as the primary revenue leader, followed closely by vegetarian anchors.

Customer "Hero" Products

Identification of high-value offerings reveals a strong preference for North Indian classics among premium members.

★ **Shahi Paneer:** Overall volume leader (8 total orders).

🏠 **Dal Makani:** Most common entry-point product for new users.

🏆 **Premium Tier:** Shahi Paneer & Dal Makani account for 15 orders within Gold users.



Loyalty Program Performance



Gold Member Impact

Total Gold Revenue: Rs. 6,890

Every registered user in this sample has transitioned to the Gold Membership tier, indicating high engagement with loyalty perks.

Customer Lifetime Value Metrics

1,510

Max Spender (Rs.)

User ID 1 (Anshul) leads contribution.

13

Loyalty Ceiling (Yrs)

Users 7 & 8 show decade-long stability.

The Actionable Roadmap



Monetize Habits

Create "Gold Platters" bundling Butter Chicken & Shahi Paneer to increase Average Order Value (AOV).



Optimize Lifecycle

Use Dal Makani as a "Welcome Voucher" for new signups to bridge the gap to Gold conversion.



Program Expansion

Introduce a "Platinum Tier" for whales (spenders > Rs. 1000) with priority prep and zero delivery fees.

Factual Understanding Over Intuition



Through intensive SQL-based analysis, we successfully transformed transactional data into a **strategic roadmap**.

- ✓ Proved loyalty as a massive revenue anchor.
- ✓ Identified specific "hook" and "value" products.
- ✓ Enabled operational readiness for non-technical teams.

Questions?

Exploring Customer Behavior Through Database
Analysis

Anjali Sinha | Data Analyst Case Study
SQL Server Management Studio | 2024 Analysis
Report

Image Sources



<https://cdn.dribbble.com/userupload/19582538/file/original-33d2ebd1cd09396d32d635fa4fdcfbb4.jpg?format=webp&resize=400x300&vertical=center>

Source: dribbble.com



<https://www.conceptdraw.com/How-To-Guide/picture/Entity-Relationship-Diagram-Software-Engineering.png>

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